Retrieval Fidelity in RAG Systems

Is the way we search for information fundamentally broken?

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Background: What is RAG?

The Purpose of RAG

- Connects LLMs to private, external, or up-to-date databases.
- Provides necessary context before the LLM generates an answer.

The Retrieval Mechanism

- Searches the database using vector mathematics.
- **The Assumption:** If the text mathematically matches the query, it contains the logical answer.

The Danger of "Semantic Noise"

What is Semantic Noise?

- Text that mathematically matches the question, but logically fails.

The Vector Illusion

User Question: *"Who conquered the Normans?"*
Retrieved Text: *"The Normans conquered England."*

Result: A perfect vector match, but logically useless context.

Dataset Setup (SQuAD 2.0)

4,000 Question-Paragraph Pairs

Example Query: "Who conquered the Normans?"

- **True Matches (2,000):** Paragraph contains the actual answer.
  - *Ex: "King Philip II of France conquered the Normans in 1204."*
- **Random Noise (1,000):** Completely unrelated text.
  - *Ex: "Photosynthesis is the process by which plants make food."*
- **Adversarial Noise (1,000):** Trick questions mimicking the query.
  - *Ex: "The Normans conquered England in 1066."*

Feature Engineering: Measuring the Relationship

Mirroring Modern Hybrid Search:

- We extract two core metrics to measure the mathematical relationship between the Question and the Context.

1. Cosine Similarity

- Measures the angle between vectors.
- Captures *semantic direction*, independent of text length.

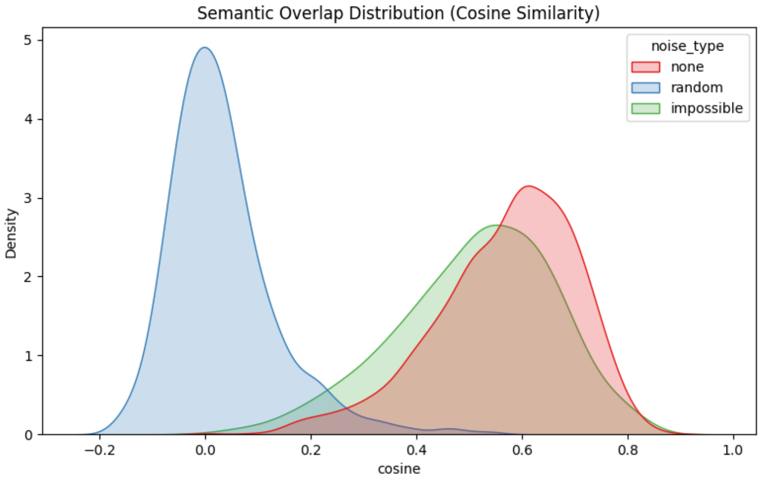
$$\cos(\theta) = \frac{q \cdot c}{\|q\|_2 \|c\|_2}$$

2. Jaccard Index

- Intersection over Union.
- Captures exact *lexical overlap* (keyword matching).

$$J(Q, C) = \frac{|Q \cap C|}{|Q \cup C|}$$

Data Distribution: Cosine Similarity

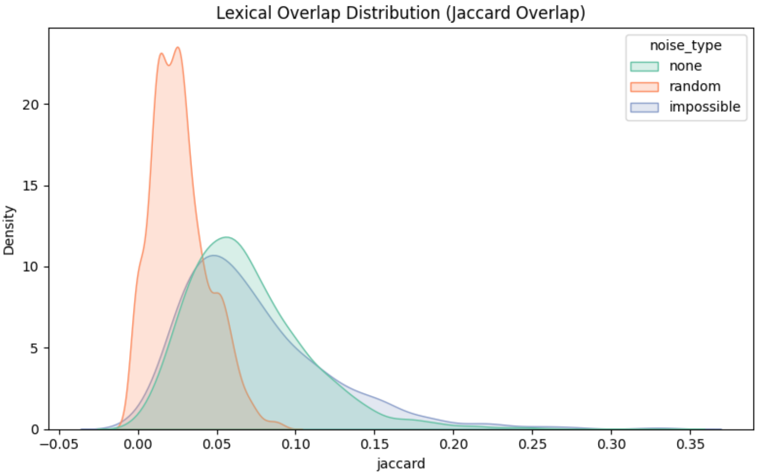


Analysis: The Semantic Overlap

Diagnostic Finding

Adversarial tricks (Green) perfectly mirror True Matches (Red). Standard vector space cannot distinguish them.

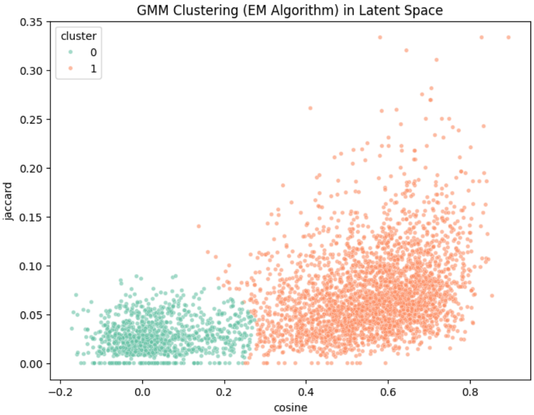
Data Distribution: Jaccard Overlap



Analysis: The Lexical Wall

- Tricks use the same keywords as the question.
- Keyword matching (Sparse retrieval) is blind to logical direction.

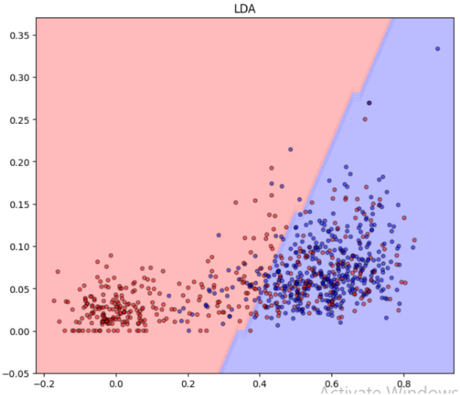
Unsupervised Test: GMM ($n = 2$)



Analysis: GMM Failure

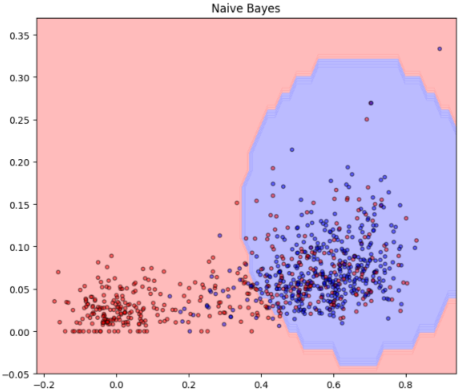
- **Cluster 0:** Successfully caught Random Noise.
- **Cluster 1:** Merged Facts and Tricks into one group.

Supervised Learning: Linear Discriminant Analysis (LDA)



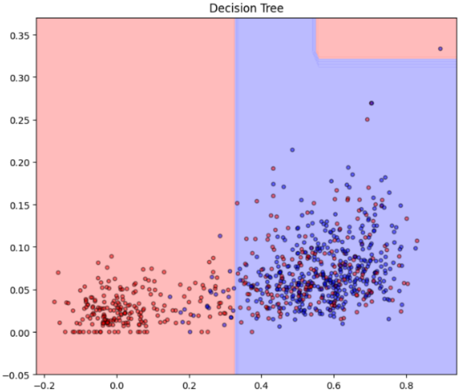
Result: Draws a straight, rigid line. Too stiff for the overlapping data.

Supervised Learning: Gaussian Naive Bayes



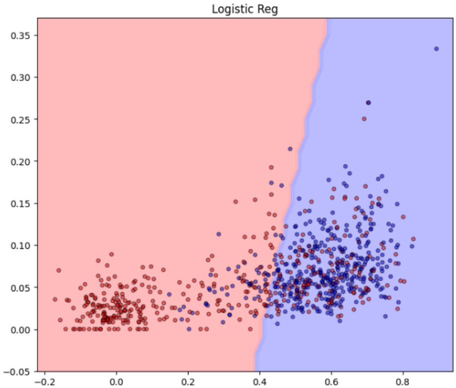
Result: Boundary warps unnaturally because it wrongly assumes the features are independent.

Supervised Learning: Decision Tree



Result: Severe overfitting. It draws tiny boxes to memorize the noise instead of learning a rule.

The Winning Model: Logistic Regression



Result: The smoothest, most reliable boundary. It balances the features without overcomplicating things.

The Performance Plateau

5-Fold Cross-Validation Metrics

Model	Accuracy	Precision	Recall	AUC
Logistic Reg	0.739	0.702	0.830	0.809
LDA	0.746	0.701	0.860	0.812
Naive Bayes	0.742	0.691	0.876	0.788
Decision Tree	0.741	0.682	0.902	0.798

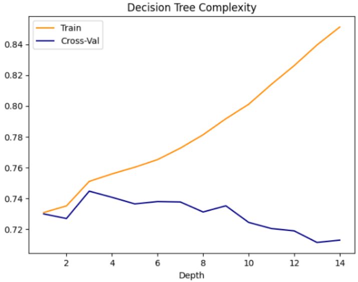
- **Accuracy:** Overall correctness (True Matches + Noise).
- **Precision:** Out of all the paragraphs it labeled "Valid," how many were actually valid?
- **Recall:** Out of all the actual "Valid" paragraphs, how many did it successfully find?
- **AUC:** The model's general ability to rank good text higher than bad text.

The "Wall"

All models plateau at roughly 74% Accuracy. The bottleneck is the **features**, not the algorithms.

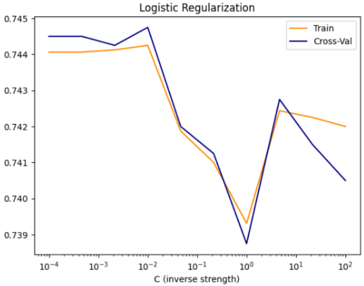
Complexity Control: Can Tuning Fix It?

Decision Tree (Depth)



Overfits past Depth 4.

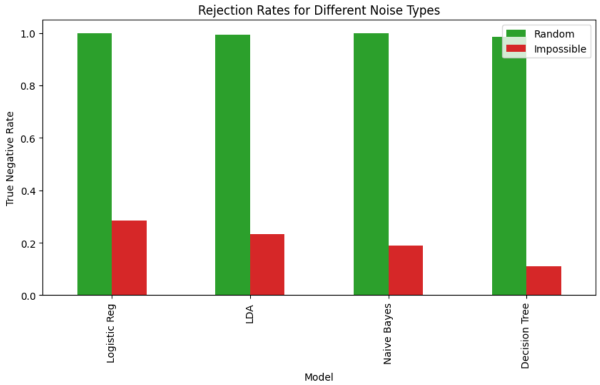
Logistic Regression (L2 Penalty)



Stays flat.

- **The Diagnosis:** Adding L2 Regularization (C) proves that "tuning" doesn't help. The Logistic Regression model has already squeezed every possible piece of information out of the vector space.

The Specificity Collapse



The Final Test

95% rejection of Random Noise vs. **28% rejection** of Adversarial Noise.

The model filters "garbage" easily, but is completely blind to logical tricks.

Conclusion

- Logistic Regression is an efficient filter for random noise, but not for "logical hallucinations."
- **Similarity is not Logic.** Vector math cannot detect semantic contradictions.
- **The Future:** The industry must move toward logic-aware systems (like Cross-Encoders) that actually read context before answering.